Wildchrokie ms

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Methods

Common garden setup and data collection

We collected seeds from 18 species of deciduous trees and shrubs from four provenances across a 3.5° latitudinal gradient (Figure 1 and Table S1) in 2014-2015, germinated them following standard germination protocols and grew them to seedlings (Baskin & Baskin, 2014). In the spring of 2017, we planted them outside in the garden located near the Weld Hill Research Building at the Arnold Arboretum in Boston MA (42.30°N, 71.13°W). Plots were regularly weeded and watered throughout the duration of the study and were pruned in the fall of 2020 (Buonuito, unpublished).

For the years 2018-2020, we monitored leaf phenology for all individuals in the common garden once or twice per week from February to December; the exact monitoring frequency varies (see Figure S1). For this, we used a modified BBCH scale to observe the phenological stages (e.g. leafout: BBCH 15 and used an additional phase for budset: BBCH 102; (Finn *et al.*, 2007); Buonuito, unpublished). We had a total of 239 leafout and 230 budset observations (which we refer as the full datasets), together yielding 227 full growing season observations (restricted dataset).

In the spring of 2023, we collected tree ring samples only for the tree species with high survivorship. Thus, we have four species in the *Betulaceae* family : Grey alder (*Alnus incana*), Yellow birch (*Betula alleghaniensis*), Paper birch (*Betula papyrifera*) and Grey birch (*Betula populifolia*). We collected cross-sections on 78 individuals by cutting the trunk above the root collar, and cores on 43 individuals using standard dendrochonological methods (Cook & Kairiukstis, 1990), for a total of 81 individuals (Table S1). We stored the cores in modified plastic straws to allow aeration and left them to dry with the cross-sections at ambient temperature for three months.

Sample processing, imaging and measuring

We mounted cores on wooden mounts, and sanded both the cores and cross-sections using progressively finer sandpaper grits (150, 300, 400, 600, 800, 1000), then scanned the cores and cross-sections at a resolution of 6250 dpi with a high-resolution treering scanner (Fong, unpublished). Using the digitized images, we measured ring widths with ImageJ (Schneider *et al.*, 2012) at three standard locations for each cross-section (at 120° from each other, averaging the three measurements to yield one ring width per year, per individual), and one ring width for each core. When we had cross-sections ($n = 78) and cores($n = 43) for an individual, we used only the measurement from the cross sections ($n = 81$).

We performed visual cross dating. This consists of assigning a calendar year to each ring and compare the year-to-year growth variation and anomalies across replicates of the same species (Cook & Kairiukstis, 1990). We did not perform statistical crossdating or detrending because our chronologies (years of data) were extremely short (Raden *et al.*, 2020) and showed no evidence of consistent allometric trends (Figure S2). We controlled for effects of different years by including year in our models (see 'Data analyses' below)

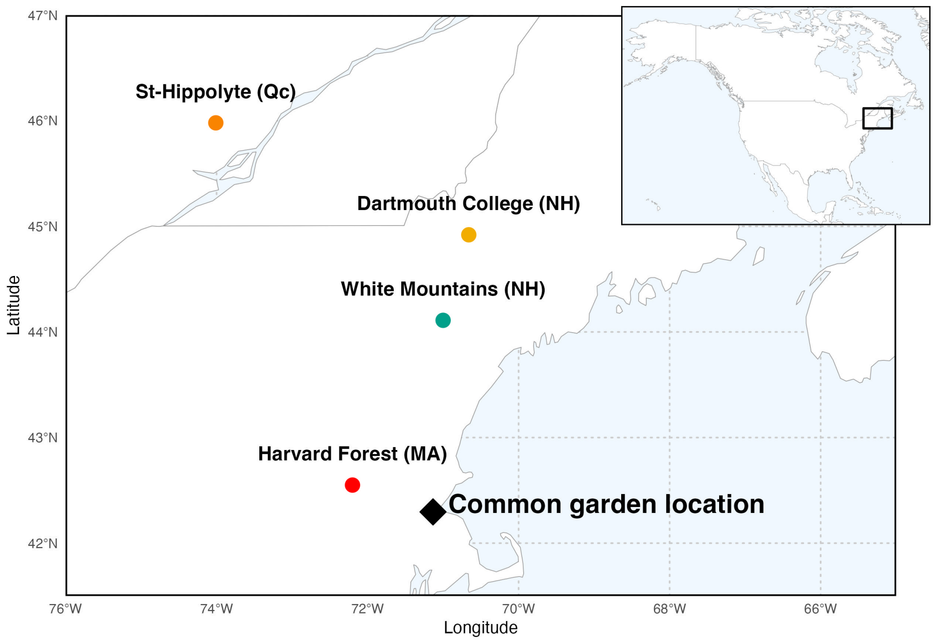

Figure 1: Provenance where the seeds were sourced from are depicted as circles, while the diamond denotes the location of the common garden. The common garden location is where the study took place at the Weld Research Building (Boston, MA). The colors of the dots match the model estimates of Figure 3.

Data analyses

To understand how growth shifted with season lengths we used linear Bayesian hierarchical models, estimating how growth, measured as ring width, covaried with season length, measured in four different season ways: GDD, GSL, SOS and EOS. We calculated the thermal season using growing degree days (GDD; mean daily temperature above 5°C) accumulated between leafout and budset, and the calendar season (or growing season length (GSL)) as the number of days between leafout and budset. Finally, we used leafout and budset as the start (SOS) and end of the season (EOS) as separate predictors that represent the boundaries of the growing season. For interpretability, we report ring width change (on the log scale) as the number of GDDs accumulated in an average week in May (day of year 121 to 150 from 2018 to 2020), for the thermal season, and ring width change per seven days of season length, leafout and budset for GSL, SOS and EOS, respectively.

We used the daily climate data from the climate station located at Weldhill research station for 2018 and 2019 (Arnold Arboretum, 2026). Because the data was not available for that station after August 2020, we used the daily climate data from the weather station "Boston (Weld Hill)" of the Network for Environment and Weather Applications (NEWA) for the whole year in 2020 (Carroll *et al.*, 2026). We used the North American Drought Monitor Maps to determine drought occurence across year at our study location and used the proposed drought intensity to qualify the moderete, severe and extreme droughts (National Centers for Environmental Information (NCEI), 2026)
For each predictor, we estimated its effect on ring width observed (i, denoting each observation) for each tree ($treeid$) in each year ($year$) as:

$$
\begin{aligned}
ln(ringwidth)_i &\sim N(\mu, \sigma_y) \\
\mu &= \alpha + \alpha_{\text{species}} + \alpha_{\text{prov}} + \alpha_{\text{treeid}} + \alpha_{\text{year}} + \beta_{species} \times \text{season predictor} \\
\alpha_{\text{prov}} &\sim N(0, \sigma_{\alpha\,\text{prov}}) \\
\alpha_{\text{treeid}} &\sim N(0, \sigma_{\alpha\,\text{treeid}})
\end{aligned}
$$

This model estimates a species-specific effect of each season length predictor (e.g. GDD) on growth ($\beta_{species}$), while also controlling for other variation due to species ($\alpha_{species}$), provenance (α_{prov}), individual tree (α_{treeid}) and year (α_{year}). We ran the models using both natural units and scaled predictors through z-scoring (Gelman & Hill, 2006) to allow for direct effect size comparison. We report results estimates as means and 90% uncertainty intervals (UI).

We used weakly informative priors for all models (increasing the priors by 2x for the z-scored models did not qualitatively affect the results). We ran four chains with 2000 warmup and 2000 sampling iterations each, for a total of 8000 sampling iterations. The models did not have any divergent transitions, had a $\hat{R}$ below 1.01, and values of effective sample size (n_{eff}) greater than 10% of the model iterations. We wrote our models using Stan (Carpenter *et al.*, 2017) with the rstan package version 2.32.7 (Stan Development Team, 2025) to run the Stan code in R.

Because ring widths decrease with age—particularly with young trees—we also ran the models using basal area increment (BAI) as the response variable. This allows for a more representative portrayal of annual growth increment by using the total ring area, instead of its width alone. For this, we summed each ring width separately at the three distinct locations on the cross sections yielding three radii from pit to bark. We then calculated the area of each ring by subtracting the radius from the inner ring from the outer ring: $\pi \times (\text{Inner radius}^2 - \text{Outer radius}^2)$, and averaged across the three BAI for each ring. Finally, we also fit the models for SOS and EOS using both the restricted (n =227) and full datasets for each predictor (SOS n =239 and EOS n =230.

Results

Our dataset includes four deciduous tree species from four sites with leaf phenology data spanning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals. At the common garden location, the mean summer temperature was similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C). There was no drought for those three years, except for in 2020, where a moderate summer drought (June, July and August) ramped up to a severe drought in September (National Centers for Environmental Information (NCEI), 2026). The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018) (Figure S8a and b). The earliest leafout was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020) (Figure S8c. The earliest budset was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018) (Figure S8d. Finally, annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018; Figure S7).

Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it. We report results as means and 90% uncertainty intervals (UI) that correspond to ring width change in mm on the log scale—henseforth "RW" per scaled predictor (7 average spring growing degree days (GDD); (55.73) for the thermal season, 7 days of growing season length (GSL) for the calendar season, 7 days of leafout for the start of season (SOS) and 7 days of budset for the end of season (EOS). Warmer thermal season increased growth for Paper birch (*Betula papyrifera*; 0.15, UI: 0.06, 0.23 RW change per scaled GDD), Grey birch (*Betula populifolia*; 0.07, UI: 0.02, 0.12 RW change per scaled GDD), and Yellow birch (*Betula alleghaniensis*; 0.04, UI: 0, 0.08 RW change per scaled GDD), Grey alder (*Alnus incana*) trended in the same direction (0.02, UI: -0.01, 0.05 RW change per scaled GDD); Figure 2a and e; TableS2). Unlike its response to warmer thermal season, *A. incana* grew more with longer calendar seasons (0.05, UI: 0. 0.1 RW change per seven GSL days). *B. papyrifera* also grew more with longer calendar seasons (0.18, UI: 0.06, 0.31 RW change per seven GSL days), while the other two *Betula* did not, though they are trending in the same same way (*B. alleghaniensis*: 0.03, UI: -0.03, 0.09 RW change per seven GSL days, and *B. populifolia*: 0.06, UI: -0.02, 0.13 RW change per seven GSL days; Figure 2b and f; Table S2). On a scaled axis (through z-scoring (Gelman & Hill, 2006)), both predictors were similarly predictive (Figure S9). Additionally, the results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S5).

Increased growth with longer seasons is often presumed to come from growth in the spring, since spring phenology has shifted much more than fall phenology. However, while we found that a longer calendar season increased growth for two species, only one also grew more with an earlier start of season (*A. incana* -0.07, UI: -0.14, -0.01 RW change per seven SOS days; Figure 2c and g; Table S2). However, *B. alleghaniensis* which did not grow more with longer calendar season, grew less with an earlier start of season (0.1, UI: 0, 0.2 RW change per seven SOS days; Figure 2c and g; Table S2)—but more with a later end of season (0.12, UI: 0.05, 0.18 RW change per seven EOS days). Alongside *B. alleghaniensis*, two other species grew more with a later end of season (*B. populifolia* 0.1, UI: 0.02, 0.17 RW change per seven EOS days, and *B. papyrifera*: 0.21, UI: 0.09, 0.33 RW change per seven EOS days; Figure 2d and h; Table S2), with the latter being the only one of those three that also grew more with longer calendar season. The estimates were unchanged when using the full (SOS $n = 239$ and EOS $n = 230$) or restricted datasets ($n = 227$; Figure S3a).

We did not find any clear effect of year on growth (2018: 0.18, UI: -0.76, 1.1 RW (on the log) deviation from the grand mean; 2019: 0.16, UI: -0.78, 1.08 RW deviation from the grand mean; 2020: -0.45, UI: -1.38, 0.48 RW deviation from the grand mean; Table S4), and there was no apparent differences of growth looking across species (*A. incana*: 1, UI: -3.08, 5.07 RW deviation from the grand mean; *B. alleghaniensis*: 0.15, UI: -3.97, 4.24 RW deviation from the grand mean; *B. papyrifera*: -4.04, UI: -8.95, 0.81 RW deviation from the grand mean; *B. populifolia*: -0.71, UI: -4.93, 3.38 RW deviation from the grand mean; Figure S4 and S5; Table S4). In addition, growth did not change across any provenances (Dartmouth College (NH): -0.03, UI: -0.23, 0.16 RW deviation from the grand mean; Harvard Forest (MA): -0.1, UI: -0.33, 0.07 RW deviation from the grand mean; St-Hippolyte (Qc): 0.04, UI: -0.15, 0.26 RW deviation from the grand mean; White Mountains (NH): 0.08, UI: -0.09, 0.3 RW deviation from the grand mean; Figure 3; Table S3). The variation across provenances ($\sigma = 0.17$, UI: 0.02, 0.47 RW deviation from the grand mean) and tree ids ($\sigma = 0.17$, UI: 0.04, 0.27 RW deviation from the grand mean; Table S4) was similar.

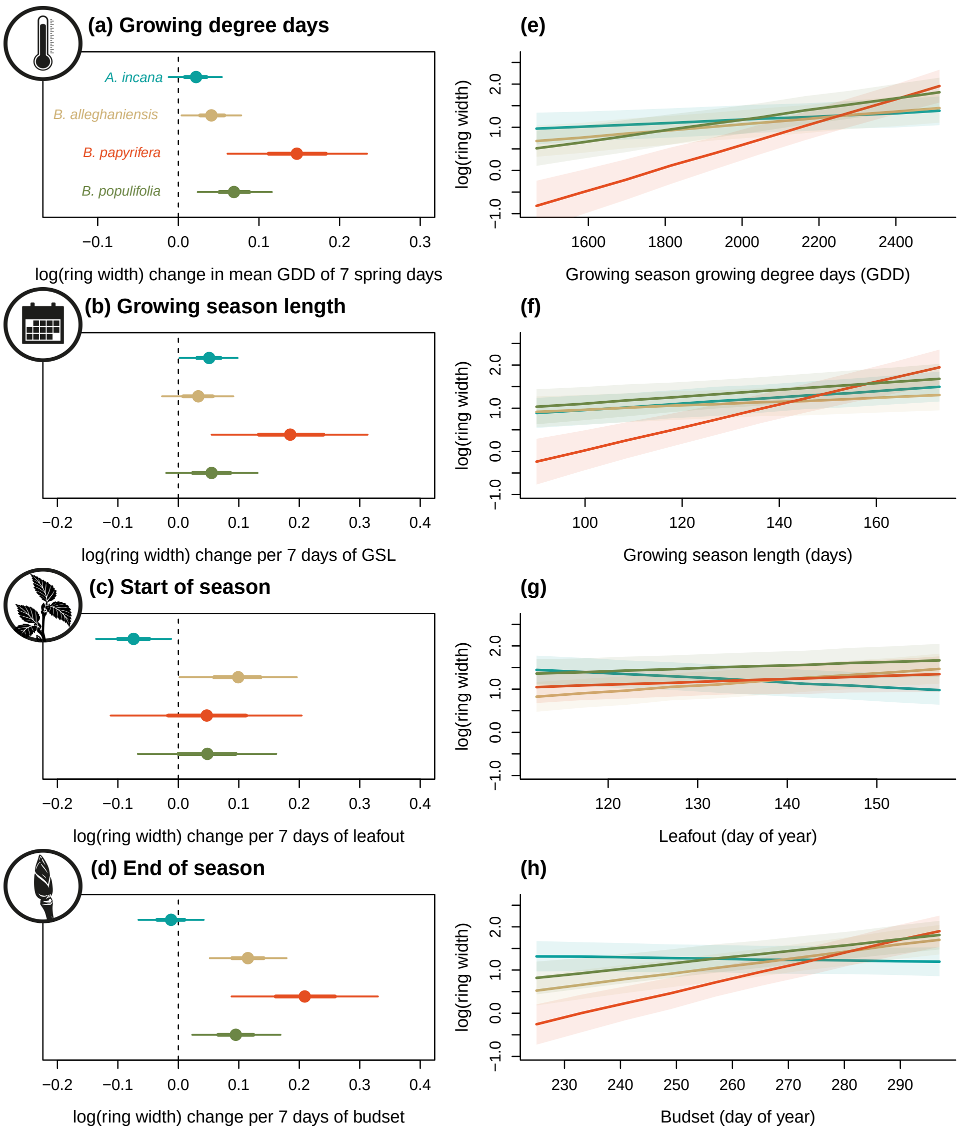

Figure 2: Estimates of ring width change on the log scale as a function of four different predictors (growing degree days, GDD (mean daily temperature above 5°C); growing season length, GSL; start of season, SOS; end of season, EOS) across four species (Grey alder (*Alnus incana*), Yellow birch (*Betula alleghaniensis*), Paper birch (*Betula papyrifera*) and Grey birch (*Betula populifolia*). The species colors are the same throughout the figures. Panels (a) to (d) represent the model slope estimates for each species, for each predictor, which we represent as ring width change on the log scale per seven average spring days GDD (55.73 GDD; panel a), seven days of GSL (panel b), seven days of leafout (SOS; panel c) and seven days of budset (EOS; panel d). The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI). See Table S2 for parameter estimates. Panels (e) to (h) represent the simulated response curves of ring width as a function of each predictor, with the lines depicting the mean response at each simulated predictor value, and the shaded area, the interquantile ranges (25% and 75% UI)
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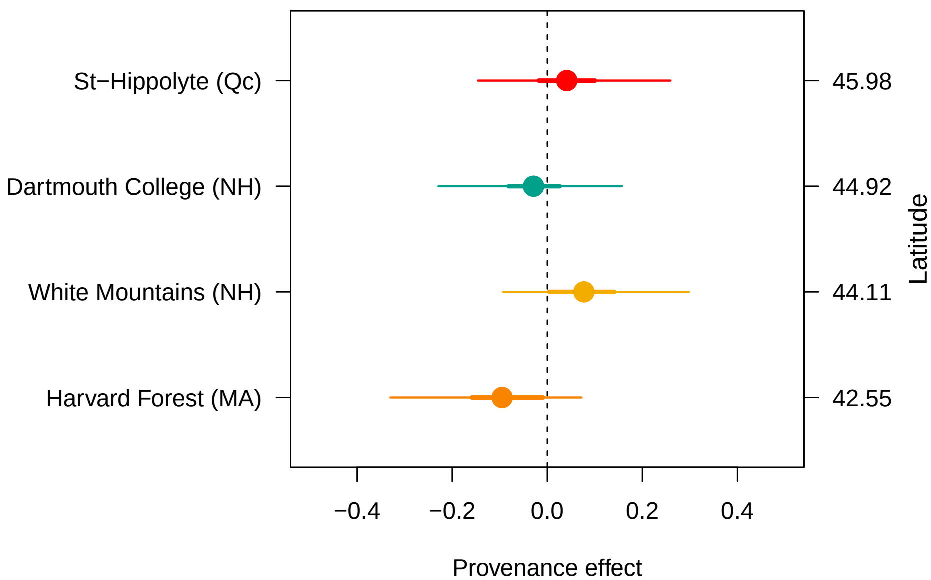

Figure 3: The effect of provenance on growth across a latitudinal gradient. We show the model intercept parameters for each provenance with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S3). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The provenances are color-coded from Figure 1, the map showing their locations.

Discussion

References

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Stan Development Team (2025) RStan: the R interface to Stan.

Supplemental material

Supplemental Methods

Table S1: Provenance coordinates and count per species and provenances out of a total of 81 individual trees.

Provenance	Longitude	Latitude	*A. incana* ($n = 29$)	*B. alleghaniensis* ($n = 20$)	*B. papyrifera* ($n = 7$)	*B. populifolia* ($n = 25$)
Harvard Forest (MA)	-72.2	42.5	7	0	4	7
White Mountains (NH)	-71.0	44.1	6	7	0	9
Dartmouth College (NH)	-70.7	44.9	8	7	2	9
St-Hippolyte (Qc)	-74.0	46.0	8	6	1	0

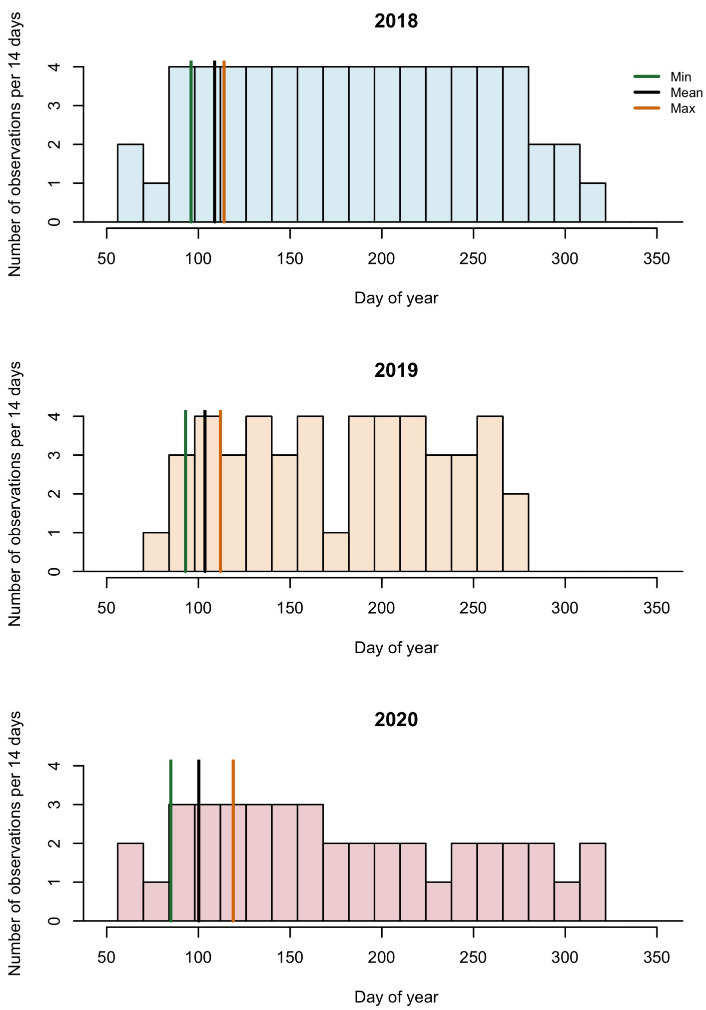

Figure S1: Leaf phenology monitoring frequency across years. We monitored leaf phenology once or twice per week for three years, though the frequency varied within and across years. Each histogram bin represent the number of observations that were made during a period of 14 days and the lines represent the earliest, the average and last recorded budburst observations. This demonstrates that despite the irregular frequency, it is unlikely that we missed early spring phenophases by more than six days.

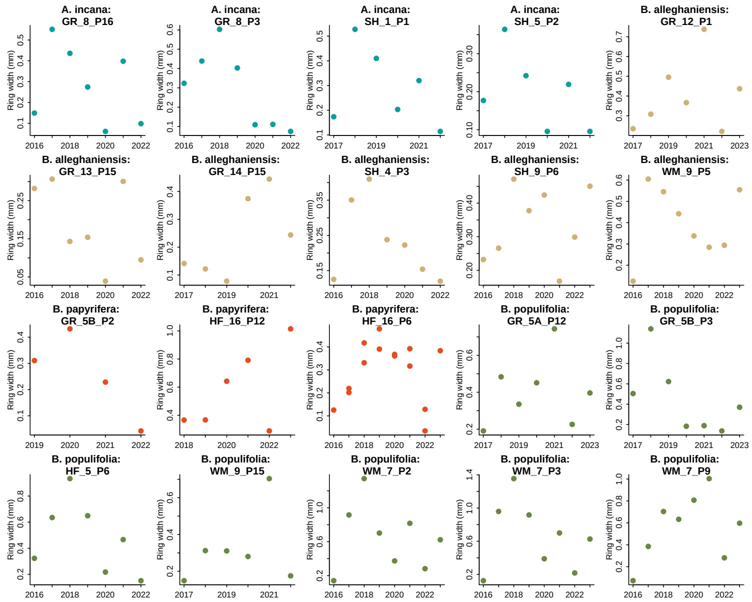

Figure S2: We randomly sampled 20 trees from our dataset and represent the ring width (mm) as a function of year to represent the allometry patterns across all years we had ring width for.

Supplemental Results

Table S2: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across our four different predictors. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.15 [0.06, 0.23]	0.18 [0.06, 0.31]	0.05 [-0.11, 0.20]	0.21 [0.09, 0.33]
B. populifolia	0.07 [0.02, 0.12]	0.06 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.02, 0.17]

Table S3: Provenance level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across our four different predictors. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Provenance	GDD	GSL	SOS	EOS	Lon.
Harvard Forest (MA)	-0.10 [-0.33, 0.07]	-0.09 [-0.35, 0.08]	-0.05 [-0.26, 0.12]	-0.09 [-0.32, 0.06]	-72.19
White Mountains (NH)	0.08 [-0.09, 0.30]	0.08 [-0.10, 0.31]	0.08 [-0.07, 0.29]	0.07 [-0.08, 0.27]	-71.23
Dartmouth College (NH)	-0.03 [-0.23, 0.16]	-0.04 [-0.25, 0.14]	-0.07 [-0.28, 0.09]	-0.04 [-0.22, 0.12]	-71.15
St-Hippolyte (Qc)	0.04 [-0.15, 0.26]	0.03 [-0.16, 0.26]	0.02 [-0.16, 0.21]	0.04 [-0.12, 0.24]	-74.03

Table S4: Posterior means and 90% uncertainty intervals for all model parameters across predictors

Parameter	GDD	GSL	SOS	EOS
$\sigma_{\alpha_{tree}}$	0.17 [0.04, 0.27]	0.18 [0.05, 0.28]	0.21 [0.10, 0.30]	0.18 [0.07, 0.28]
$\sigma_{\alpha_{prov}}$	0.17 [0.02, 0.47]	0.18 [0.02, 0.52]	0.16 [0.02, 0.45]	0.17 [0.02, 0.44]
σ_y	0.48 [0.43, 0.52]	0.48 [0.44, 0.53]	0.47 [0.43, 0.52]	0.47 [0.42, 0.52]
$\beta_{spp_{A.incana}}$	0.02 [-0.01, 0.05]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
$\beta_{spp_{B.alleghaniensis}}$	0.04 [0.00, 0.08]	0.03 [-0.03, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
$\beta_{spp_{B.papyrifera}}$	0.15 [0.06, 0.23]	0.18 [0.06, 0.31]	0.05 [-0.11, 0.20]	0.21 [0.09, 0.33]
$\beta_{spp_{B.populifolia}}$	0.07 [0.02, 0.12]	0.06 [-0.02, 0.13]	0.05 [-0.07, 0.16]	0.10 [0.02, 0.17]
$\alpha_{prov_{DartmouthCollege(NH)}}$	-0.03 [-0.23, 0.16]	-0.04 [-0.25, 0.14]	-0.07 [-0.28, 0.09]	-0.04 [-0.22, 0.12]
$\alpha_{prov_{HarvardForest(MA)}}$	-0.10 [-0.33, 0.07]	-0.09 [-0.35, 0.08]	-0.05 [-0.26, 0.12]	-0.09 [-0.32, 0.06]
$\alpha_{prov_{St-Hippolyte(Qc)}}$	0.04 [-0.15, 0.26]	0.03 [-0.16, 0.26]	0.02 [-0.16, 0.21]	0.04 [-0.12, 0.24]
$\alpha_{prov_{WhiteMountains(NH)}}$	0.08 [-0.09, 0.30]	0.08 [-0.10, 0.31]	0.08 [-0.07, 0.29]	0.07 [-0.08, 0.27]
$\alpha_{year_{2018}}$	0.18 [-0.76, 1.10]	0.22 [-0.73, 1.17]	0.21 [-0.75, 1.16]	0.23 [-0.71, 1.17]
$\alpha_{year_{2019}}$	0.16 [-0.78, 1.08]	0.09 [-0.87, 1.04]	0.11 [-0.85, 1.06]	0.12 [-0.83, 1.05]
$\alpha_{year_{2020}}$	-0.45 [-1.38, 0.48]	-0.36 [-1.32, 0.59]	-0.37 [-1.33, 0.59]	-0.47 [-1.41, 0.47]
$\alpha_{spp_{A.incana}}$	1.00 [-3.08, 5.07]	0.15 [-3.89, 4.32]	1.57 [-1.99, 5.15]	3.12 [-1.35, 7.53]
$\alpha_{spp_{B.alleghaniensis}}$	0.15 [-3.97, 4.24]	0.38 [-3.72, 4.63]	-1.84 [-5.53, 1.75]	-1.80 [-6.37, 2.65]
$\alpha_{spp_{B.papyrifera}}$	-4.04 [-8.95, 0.81]	-2.67 [-7.20, 1.95]	-0.74 [-4.90, 3.44]	-5.51 [-11.12, 0.08]
$\alpha_{spp_{B.populifolia}}$	-0.71 [-4.93, 3.38]	0.24 [-3.98, 4.52]	-0.48 [-4.34, 3.21]	-0.83 [-5.53, 3.95]

Table S5: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{prov}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{spp_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{spp_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{spp_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{spp_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{spp_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{spp_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{spp_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{spp_{B.populifolia}}$	-0.07 [-5.98, 5.57]

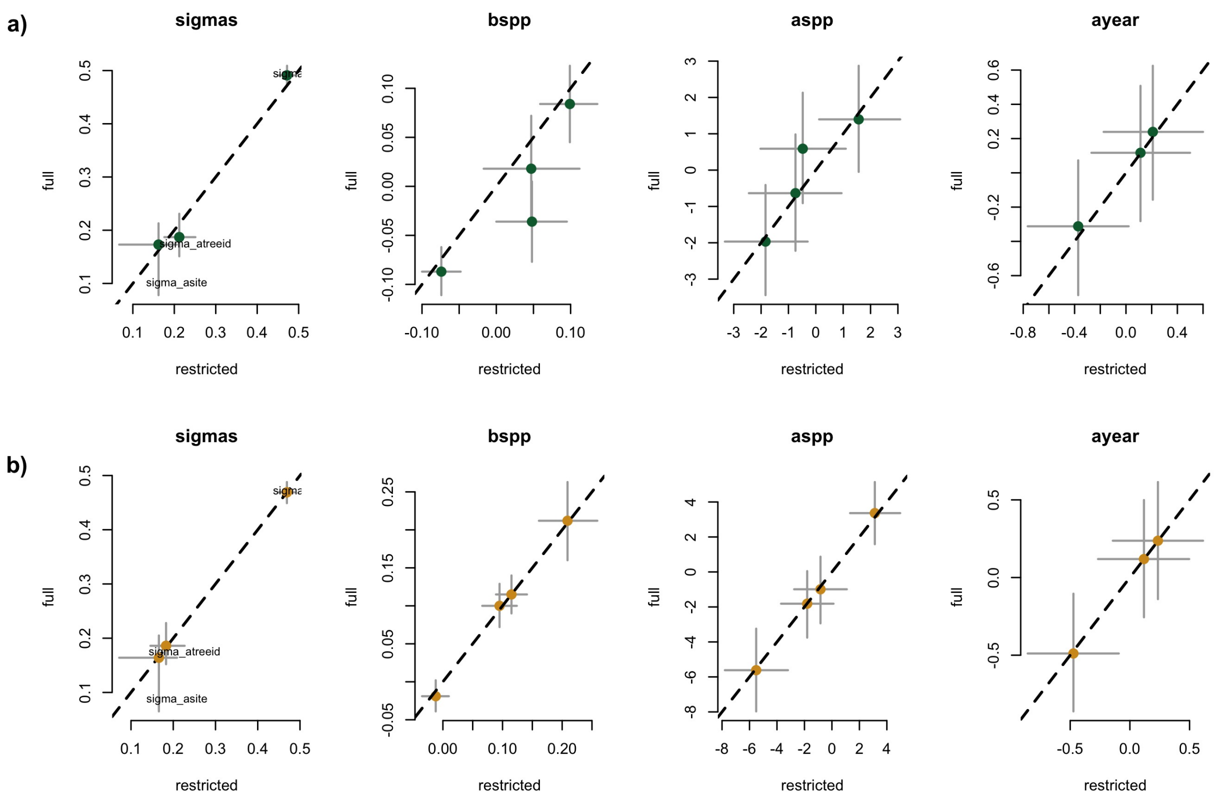

Figure S3: Full vs most restricted rows of data for the model with SOS (a) EOS (b) as the predictor. Each pannel represent the different parameters used to fit the models.

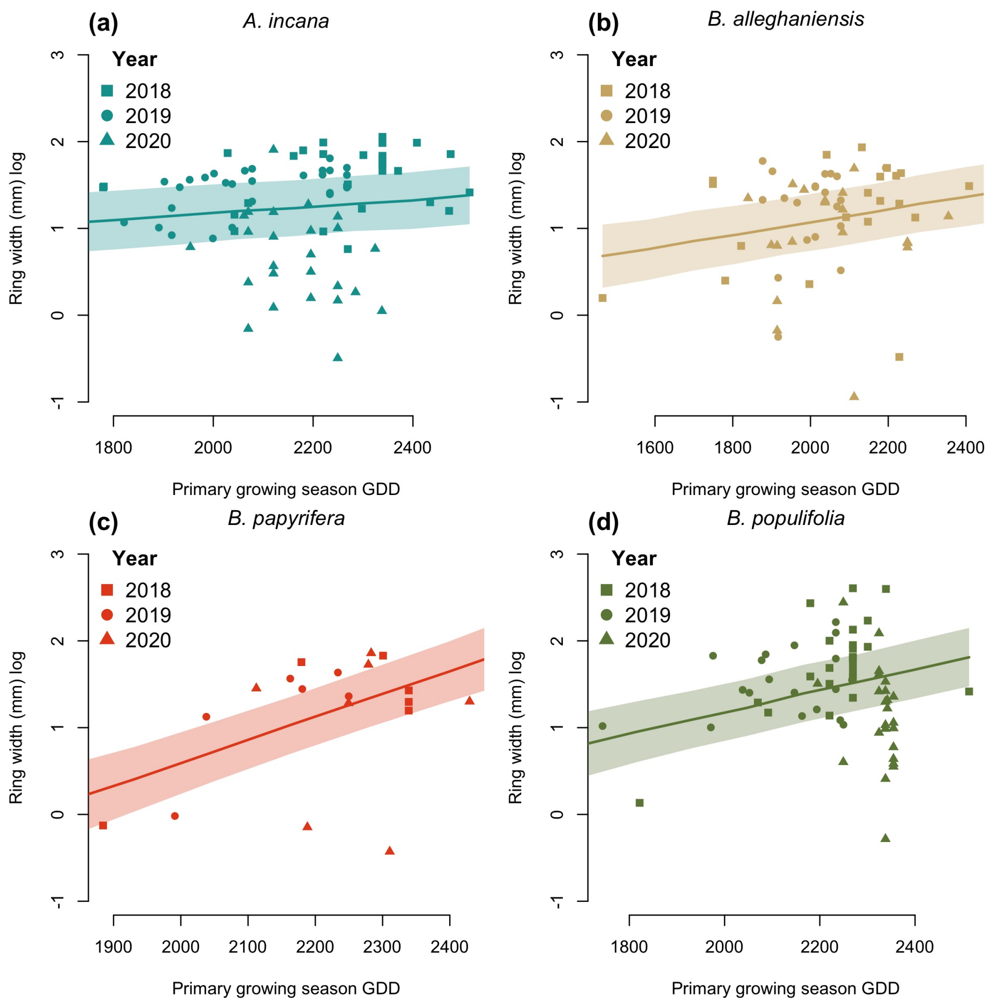

Figure S4: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

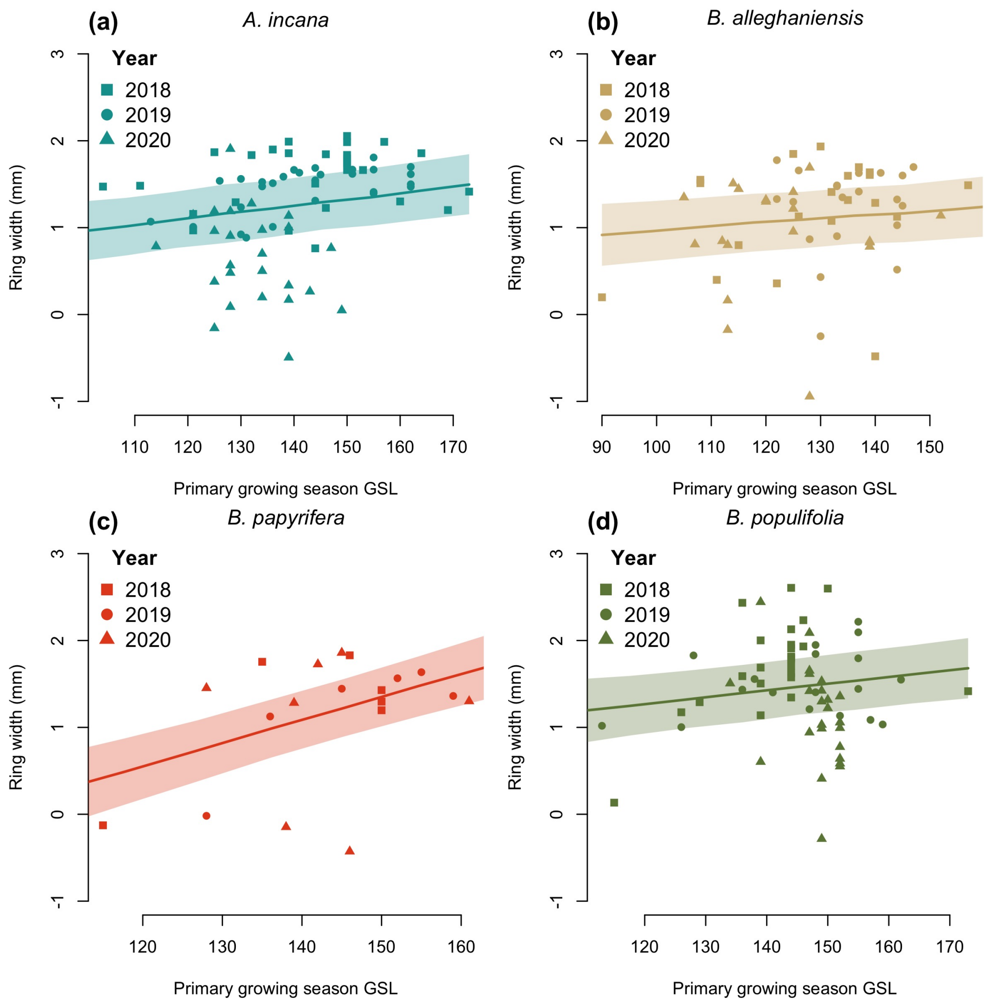

Figure S5: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the interquantile range (IQR). The dots are the empirical data shaped by year.

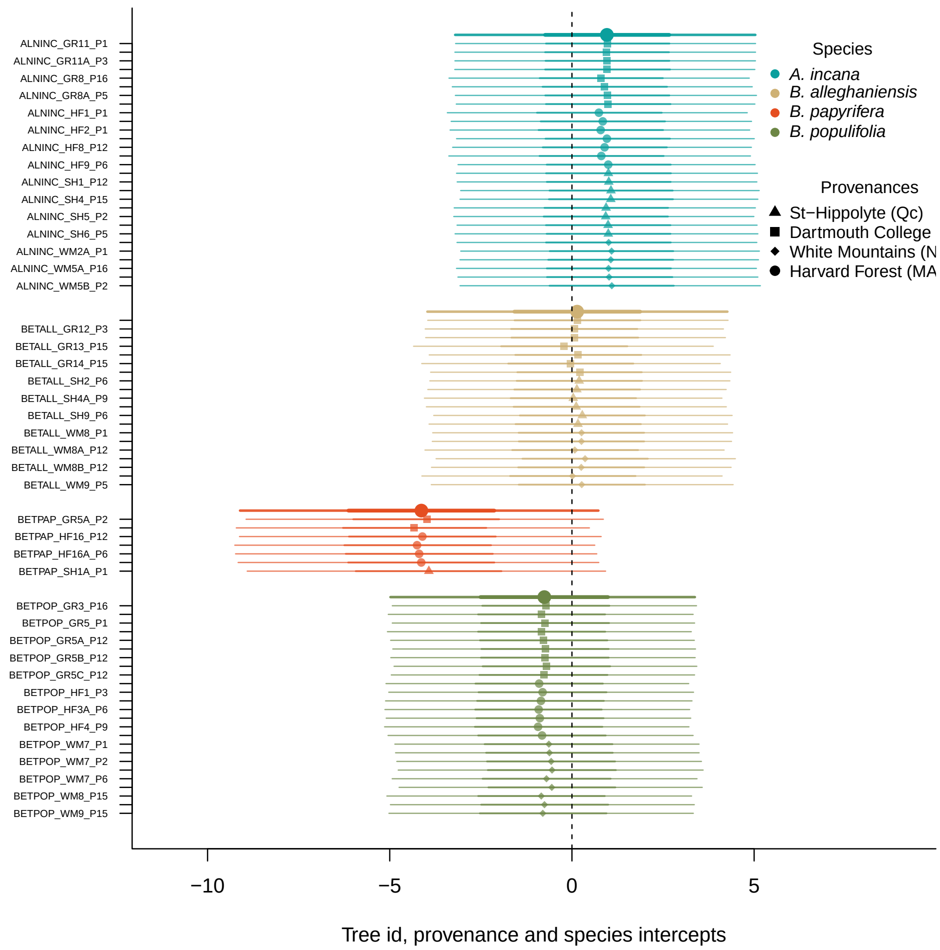

Figure S6: Intercept estimates for each group of the GDD model. The different tree id values represent the combined intercepts of tree id, provenance, species and averaged year intercepts. The bigger dots and lines represent the species intercepts which are the species level estimates averaged across all of their tree ids. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

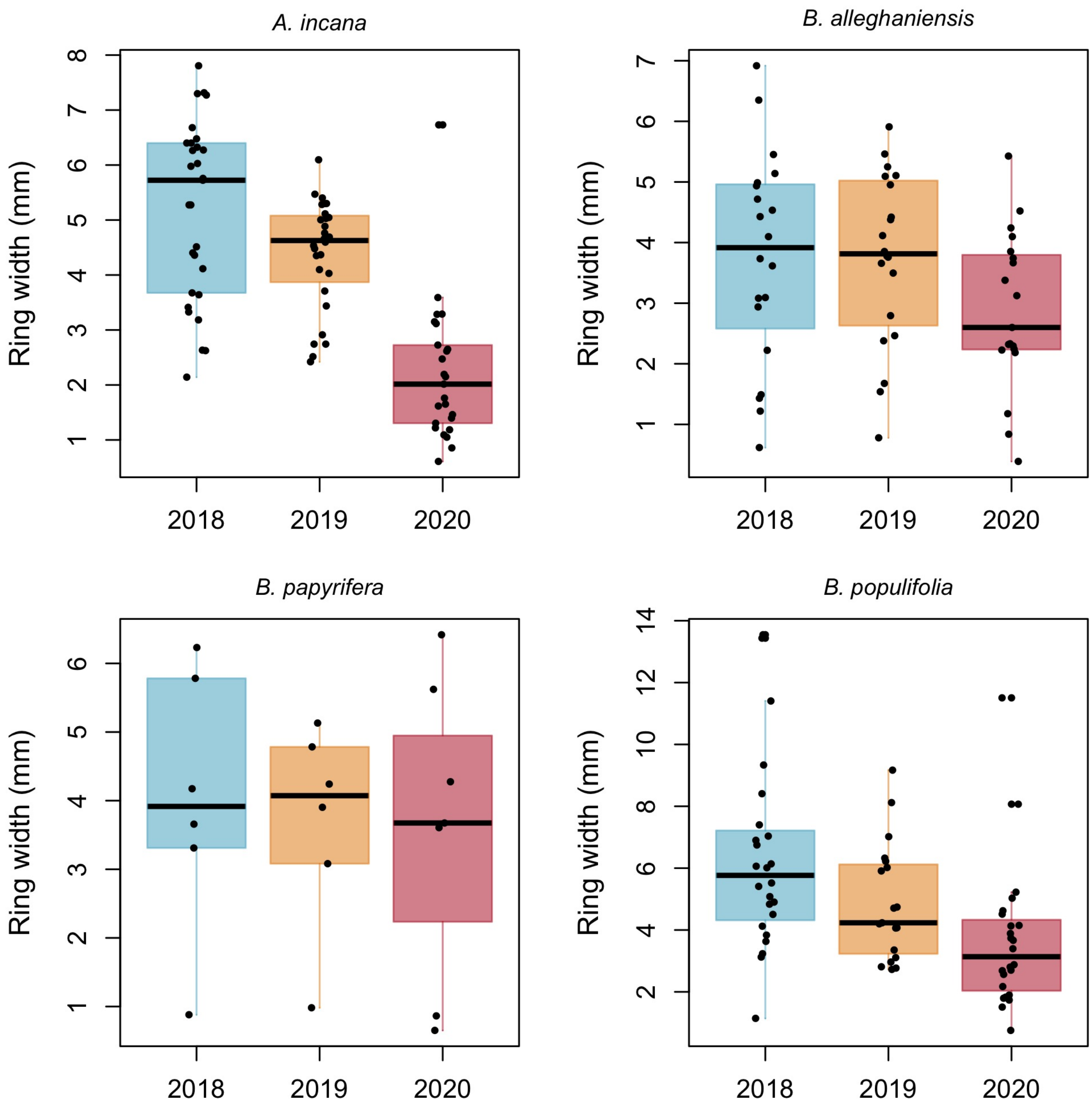

Figure S7: Boxplot representing the empirical data of our ring widths (mm) facetted by species. The dots represent the raw data, the line the mean and the box the interquantile range (IQR).

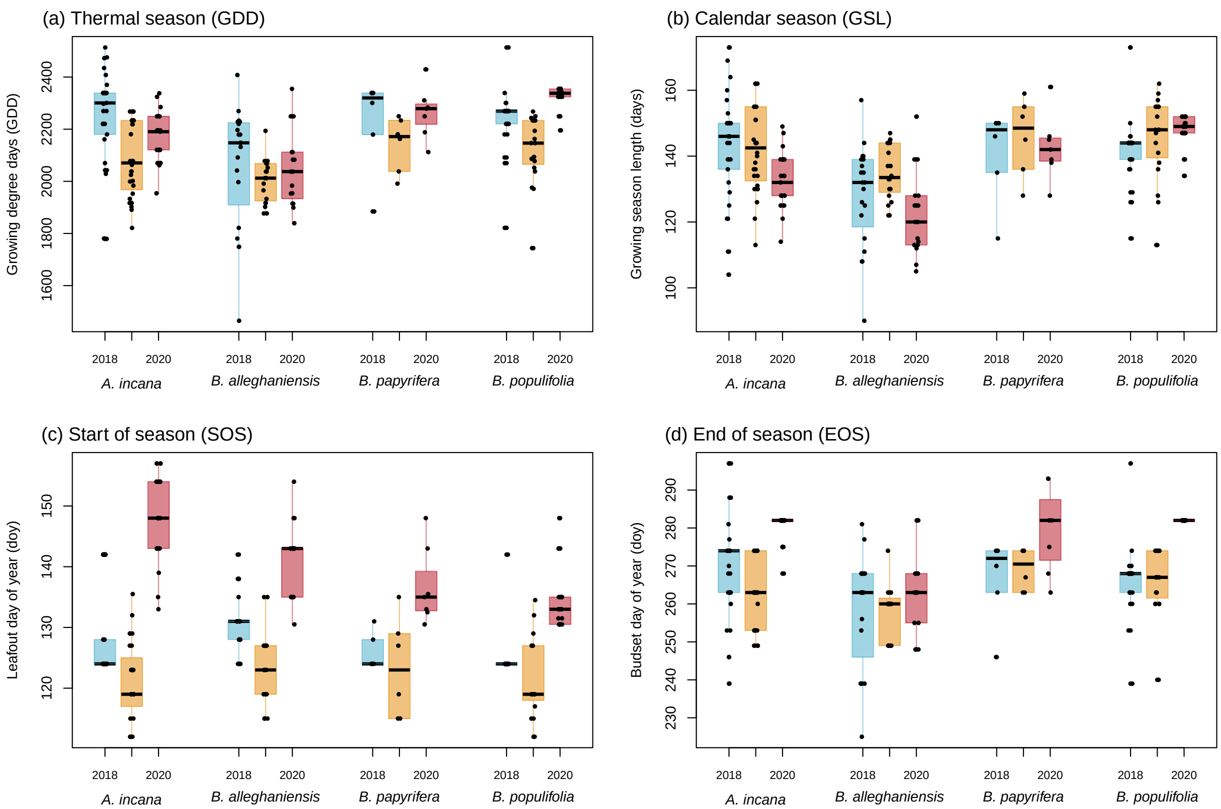

Figure S8: Boxplot representing the empirical data of all out predictors clustered by species, and for each year. The dots represent the raw data, the line the mean and the box the interquantile range (IQR).

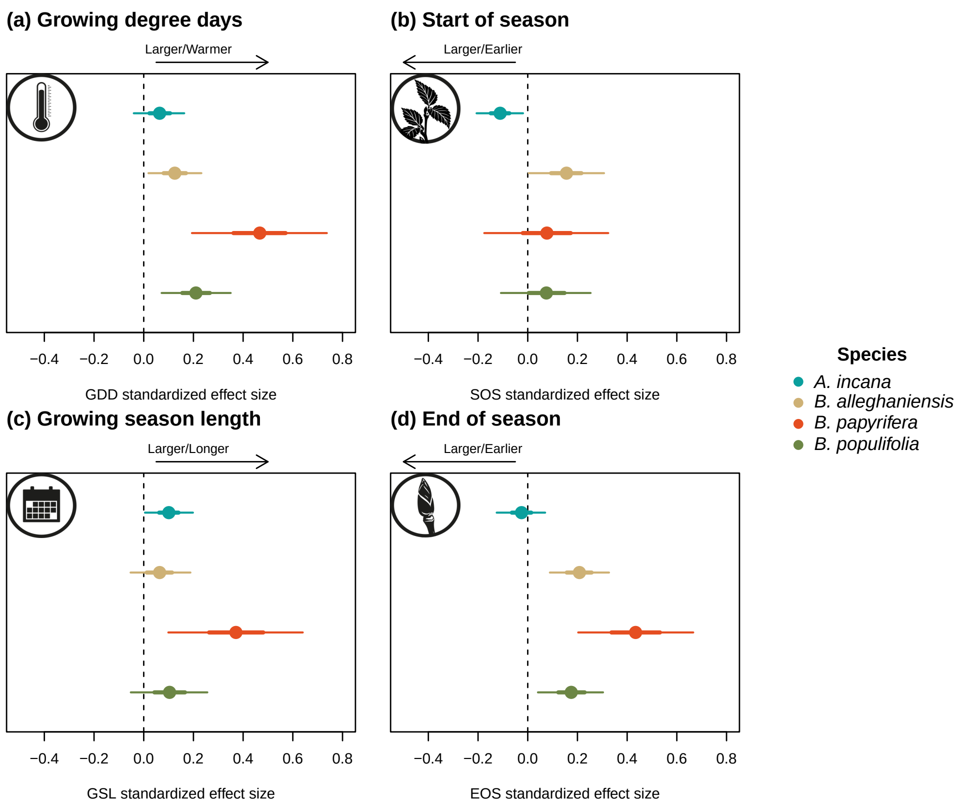

Figure S9: Standardized (z-scored) slope estimates across our four different predictors. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.